



Tool for Track Mounting in D9, D10 & D11 TTT Series on Maintenance Conveyor

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

The conveyor for the track disassembly, repair and assembly in the undercarriage shop is one of the most important steps for the proper development of these operations. It is relevant to keep this equipment in good condition and operating well. It is 15.3 meters long and has a platform made of 120 rollers which can carry all TTT series ranging from 2.1 ton to 7.2 ton in the D11 series, therefore it uses a wincher P/N 10-8.0-27-C-XL10 having enough capacity to support all the tracks. However, when using a pin latch bolt as an engaging mechanism between the two links, this has no fixed position. The wire travels thru the length of the bolt losing direction when the track moves. It also supports all the pull load of the wincher. Over time, it has significant bending due to stress. The most critical problem in this system is the teeth at the ends of the track, when they slide over the conveyor, are turned forward and got stuck in the rollers platform. This causes damage and forces the technician to use a lever and waste time to dig them back up and the track continues to slide over the conveyor. (See Fig. 1.)

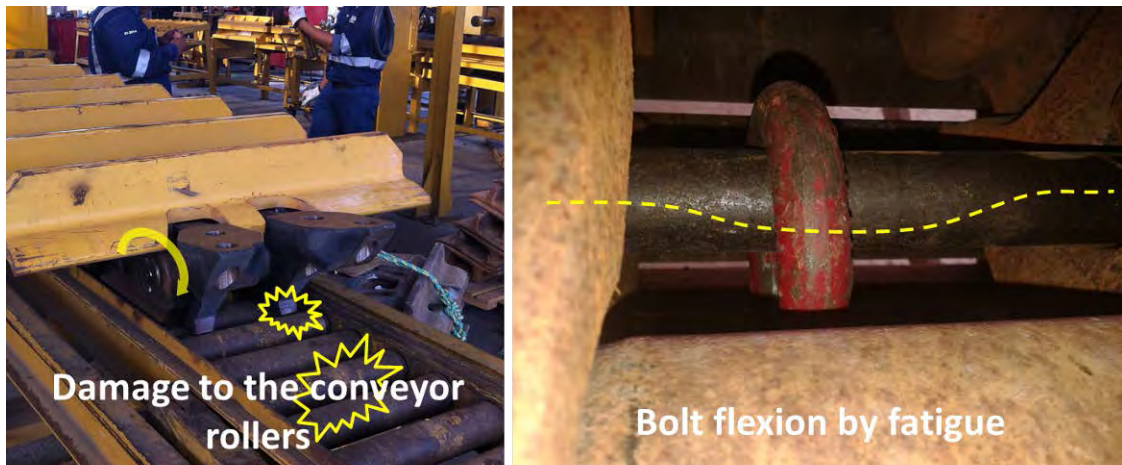


Figure 1. Problems when mounting the track on the maintenance conveyor.

This Best Practice focuses on correcting these issues, with the design of a track engaging tool (D9, D10 and D11 TTT series), that facilitates and speeds up the operation, preserving the general welfare of the rollers platform and the technicians.

2.0 Best Practice Description

This tool applies for D9, D10, D11 TTT series tracks and consists of a hook made out of a 1.5 inch steel plate (ASTM A514 Grade A) with a 3-inch wide plate of the same material that runs along the inside perimeter of the hook increasing the contact area when holding the track by the joint bushing between the links, causing the stresses generated by the wincher to be better distributed. The front end has a 1.5 inch bore for the installation of shackles that bind the tool with the winching cable (see capacities in table 2). Right above this bore, a ¾ inch x 14 inch tube is welded (Xstrong S80 ASTM A53) which intersects perpendicularly to the tool body and its ends protruding symmetrically on either side of the tool. The function of this tube is to hold the loosened teeth of the ends of the track to be mounted. These teeth lie on the tube; so they cannot rotate and damage the conveyor rollers platform.

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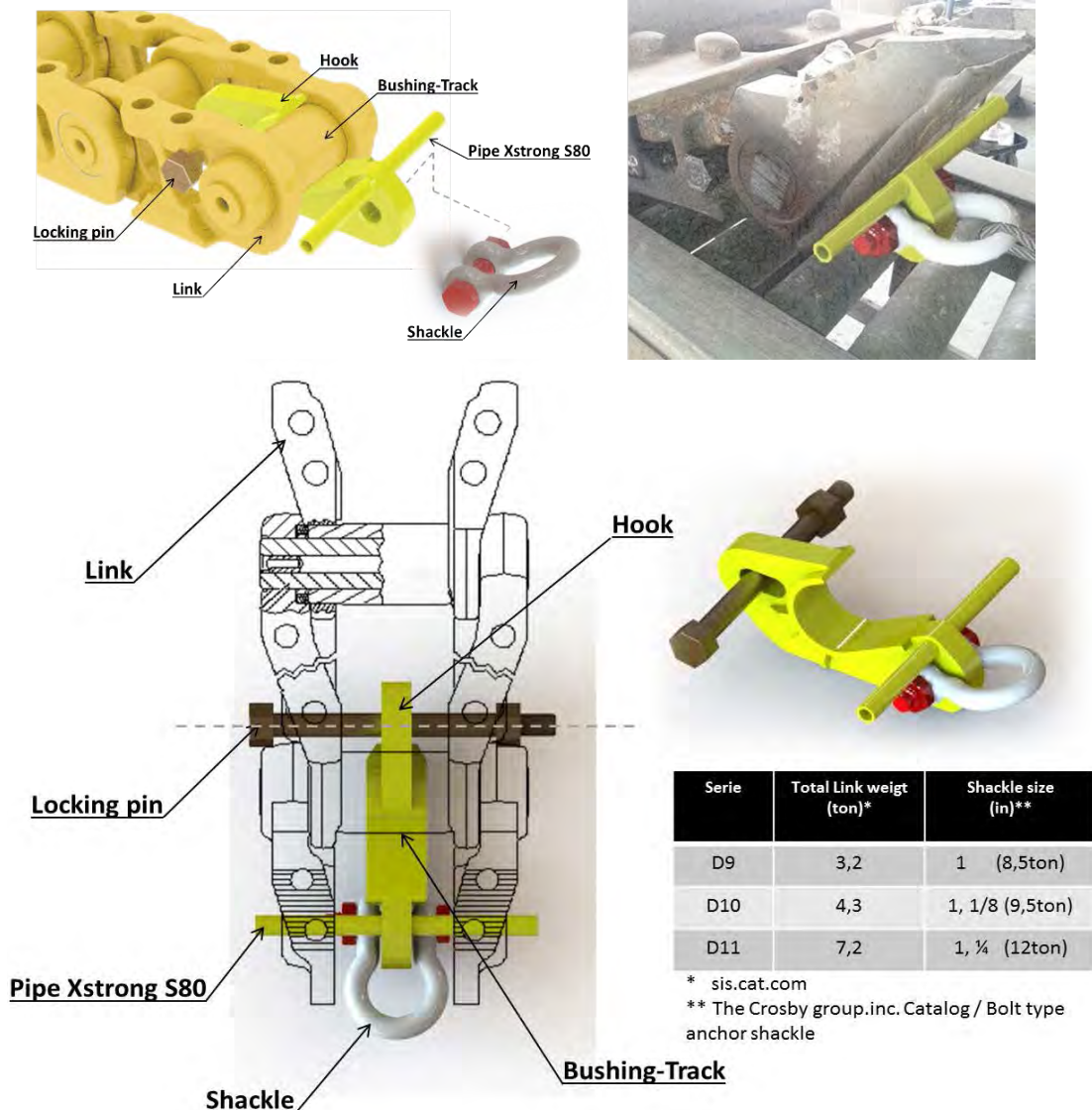


Fig. 2. Tool installation architecture.

The tool also consists of a 1 1/4inch x 12inch x 15inch bolt (P/N: 5P8335) (nut P/N: 7H7539) to keep the hook from turning on the track bushing when being pulled by the wincher. The bolt goes thru the holes of the two links and a slot strategically located in the body of the tool. The length of this groove is what allows to use it in various TTT models aforementioned, while fulfilling its function.

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3.0 Implementation Steps

- Make the tool as designed (see Fig. 2 and 4)
- Install the tool according to Fig. 2 by following these steps (see Fig. 2):
 - Initially install the shackle to be used according to the series of the track to be mounted (see table in Figure 2). It should be noted that for safety reasons we have selected a anchor type shackle with a pin having a nominal capacity of 1.5 times of the total load value, and that its opening is enough to fit in over the sheet thickness of the tool body. Once the shackle is installed, attach the tool to the bushing of the links connecting the end of the track, making sure the loose teeth of the ends are resting above the $\frac{3}{4}$ inch tube of the tool.
 - Install the locking pin and its nut. It should go right in the tool slot located behind the hook to prevent its rotation.
 - Once the tool is fully installed, proceed to operate the wincher slowly. It is important to mention that for the tool design test load, the total weight of the track was taken. However during the process, the real load is about 1/3 or more than 1/2 of this, due to the fact that the track is being mounted slowly on the conveyor and sometimes a forklift is used in order to spread it out of the conveyor.

4.0 Benefits

- Tool is easy to be manufactured, approximate cost USD \$720. It speeds up the mounting time of the track to avoid rotation of the teeth at the ends of it, not needing the technicians help. It also eliminates the risk of an accident by preventing the use a lever to lift the teeth again.
- It improves the targeting of the movement of the track on the conveyor, and it prevents damage to the rollers.
- Tool designed for better stress distribution, significantly reducing fatigue due to failures.
- Applicable to and operates correctly for several sets of tracks (D9, D10, D11).

5.0 Resources Required

- Materials and required components (see Figure 2)
- Fabrication Drawings (see Figure 4)
- \$720 USD including attachments (shackle and locking pin)

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6.0 Supporting Attachments / References

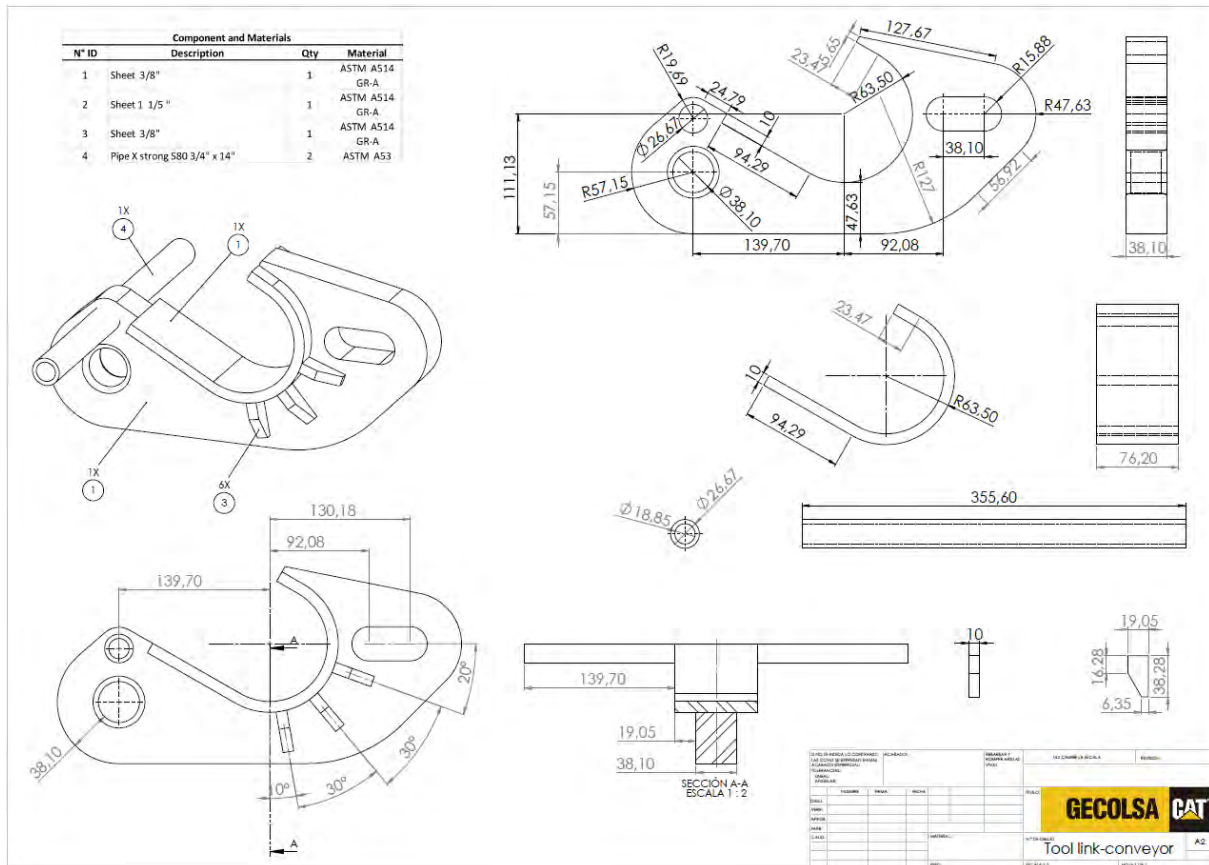
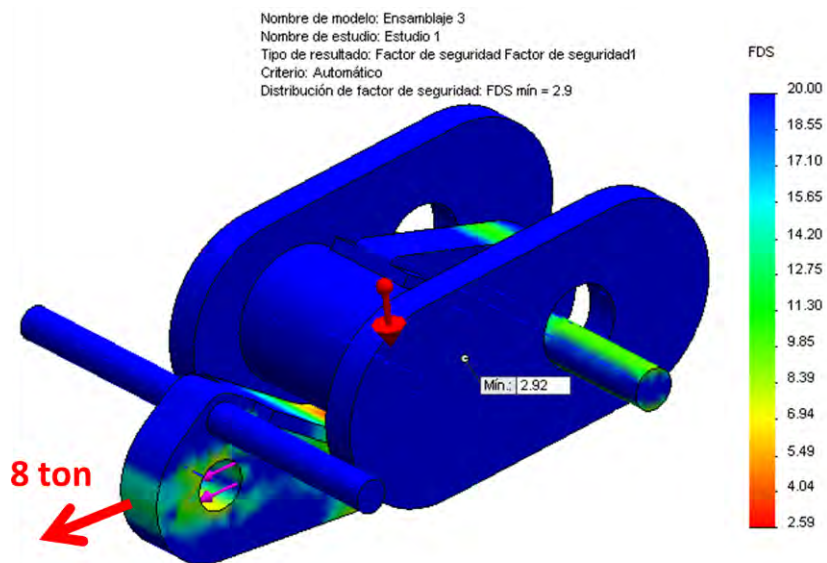


Figure 4. Tool drawing



Finite element analysis; FS = 2.9 using 1.5 times the actual workload

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7.0 Related Best Practices

None

8.0 Acknowledgements

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